

Class Imbalance in Histopathology: A Confusion-Separation Score Does Not Rank Class Orders by Damage on TCGA-UT

Yannik Queisler

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Abstract

On TCGA-UT, the damage a long-tailed training allocation does to balanced accuracy depends on which class sits at which rank, and the companion tail-assignment study traced that dependence to pairs of classes the balanced classifier confuses with each other: a class loses recall when a class it is confused with holds a larger allocation. That study named the natural worst case, an assignment that separates every strongly confused pair and places the member with more recall to lose underneath, but did not fit it. This report defines that assignment through a directed pair-separation score, which weights each ordered pair by the confusion mass flowing from one class to the other, by the first class’s balanced-arm recall, and by the log-count gap between their ranks. Before spending any of the 120 fits that a search over class orders would require, the score is checked against three damages already observed: the easy-tail order, the hard-tail order, and the average over random orders. The score fails that check. It separates the two sorted orders in the observed direction, scoring the easy-tail order at 3,239 against 895 for the hard-tail order, but it places the easy-tail order above all 30 random orders, whose scores span 1,976 to 3,043, while the easy-tail order costs 2.41 points of balanced accuracy against 2.91 points for the random orders on average. The inversion has a structural cause: the sorted orders are, by construction, monotone in the recall term the score uses as a weight, so ordering the classes by difficulty raises the score whether or not it raises the damage. Within the random orders alone the score does correlate with damage, at a Spearman coefficient of 0.38, but that correlation does not extend to the orders a search would actually visit. The search was therefore not run, and the worst assignment on TCGA-UT remains unfitted.

1 Motivation

The TCGA-UT prevalence study of this series measured how balanced accuracy on 30 cancer types degrades as the training class distribution moves from balanced to long-tailed, at twenty patients per class and 19,200 training patches held fixed (Queisler, 2026c). At an imbalance ratio of $\rho = 100$ the damage was 2.91 points, averaged over class orders drawn at random per draw. The prior-versus-support study split that damage into a part from the skewed class prior, a part from thin tail support, and a sizeable interaction between them (Queisler, 2026a).

How much of the damage depends on *which* class is rare was then tested directly. On BRACS the dependence was large and followed the *headroom* of each class, its recall under balanced training, because nearly every BRACS class collapsed to almost no recall in the tail and therefore cost whatever it had to lose (Queisler, 2026d). On TCGA-UT the picture was different (Queisler, 2026e). Sorting the classes by balanced-arm recall and placing the easiest in the tail cost 2.41 points, against 1.59 points for the reverse order, a difference of 0.82 points that carries the sign headroom predicts but stays below the one-point threshold of this series. More striking, both sorted orders cost *less* than a random order, by 0.51 and 1.33 points. No class collapsed: thyroid carcinoma kept 90 percent recall on 28 patches. What moved a class’s

recall was its allocation relative to the classes it is confused with. Rectum adenocarcinoma lost 13 points of recall while sitting just below colon adenocarcinoma, and gained 7 points while sitting above it far below balance; head and neck squamous cell carcinoma behaved the same way against lung squamous cell carcinoma, and the two gliomas, which stayed at adjacent ranks, barely moved.

That study closed with an explicit open item. If damage is set by pairs rather than by single classes, then the worst assignment should be the one that separates strongly confused pairs as far as possible in rank, with the member that has more recall to lose placed at the smaller allocation. It named this candidate and stated that it did not fit it. This report asks whether that candidate can be made precise as a score over class orders and whether the order that maximizes it is, in fact, the most damaging one. The design of that test is the subject of Section 2; the answer turns on a check that runs before any new fit and is the subject of Section 3.

2 Design

The study reuses the design of the TCGA-UT prevalence curve (Queisler, 2026c): 30 cancer types with frozen Virchow2 features (Zimmermann et al., 2024), three patient-disjoint splits with ten draws each, $G = 20$ patients per class, a total budget of $T = 19,200$ training patches, and nested patch orderings within each patient. The balanced arm r1 gives each class 640 patches. The ratio arm $r\rho$ gives the class at rank j a count proportional to $\rho^{-j/29}$, so at $\rho = 100$ the counts fall from 2,843 patches at rank 0 to 28 patches at rank 29. Only the assignment of classes to ranks is varied; patients, patch orderings, and the per-rank counts are held fixed, which is what makes any difference in damage attributable to the assignment alone.

2.1 A directed pair-separation score

Each rank r carries a log-count deviation from balance,

$$z_r = \log_2 \frac{n_r}{640}, \quad (1)$$

where n_r is the patch count at rank r ; at $\rho = 100$ these run from 2.15 at the head to -4.51 at the tail. An assignment π places class c at rank r_c . The score is

$$S(\pi) = \sum_{c \neq d} w_{c \rightarrow d} h_c \max(z_{r_d} - z_{r_c}, 0), \quad (2)$$

where $w_{c \rightarrow d}$ is the share of class c 's off-diagonal test predictions in the balanced arm that land on class d , so that each row sums to one over the other classes; h_c is class c 's pooled balanced-arm recall, standing for the recall it has available to lose; and the final factor is the log-count advantage that d holds over c , counted only when d has the larger allocation. Each term thus reads as the exposure of class c to a class it is confused with that sits above it in allocation, weighted by how much c has to lose. Maximizing S separates confused pairs and puts the member with more to lose underneath; minimizing it keeps confused pairs adjacent in rank.

Two properties of Equation (2) shaped this form. First, the score must be *directed*. Because the per-rank counts are log-linear in rank, the gap $|z_r - z_s|$ depends only on $|r - s|$, so any undirected measure of how far confused pairs are separated is unchanged when an order is reversed. The easy-tail and hard-tail orders are exact reversals of one another, so an undirected measure could not distinguish them and could not explain their observed 0.82-point difference. The one-sided max and the asymmetric weights $w_{c \rightarrow d} h_c$ supply that direction. Second, the score is rank-based, and rescaling z by a constant rescales S without changing which order maximizes it, so the same order is the maximizer at every ratio. This is what would have allowed a single search to serve both ratios.

2.2 The search and the arms it would have produced

Maximizing and minimizing Equation (2) over the $30!$ possible assignments is done by pairwise-swap hill climbing from 50 seeded random starts, taking the best improving swap at each step until no swap improves. Any result of such a search is *an* extreme assignment found by local search, not a proven optimum, and would have been reported as such.

Three orders follow from it. The order *sep* maximizes S and is the worst-case candidate. The order *tog* minimizes it and is the mildest candidate. The order *flip* reverses *sep*; because reversal preserves every pairwise rank gap, *flip* separates confused pairs exactly as far as *sep* does but exchanges the roles within each pair, which isolates the directional part of the claim from the separation itself. The planned fits were *sep*, *flip*, and *tog* at $\rho = 100$ and *sep* again at $\rho = 10$, at 30 fits each, for 120 fits in total. The r1, r10, r100, and N arms of the prevalence curve and the two sorted orders of the tail-assignment study would have been reused unchanged, as held-out points.

3 Evaluation

The endpoints of a fitted study would have been the damage of *sep* against the random-order average and the full range spanned by *sep* against *tog*, both judged against the one-point threshold of the series, with pooling and uncertainty as in the prevalence curve: equal weight per split, Bayesian bootstrap weights on test patients (Rubin, 1981) combined with resampling of draws within splits, and 95 percent percentile intervals of every paired contrast.

Those endpoints are only worth fitting if the score ranks assignments by damage in the first place. Three damages are already known and cost nothing to check against: the easy-tail order at 2.41 points, the hard-tail order at 1.59 points, and the average over the 30 random orders at 2.91 points (Queisler, 2026e,c). A gate over these three damages was fixed before the score was computed and before any fit was submitted:

- **G1:** $S(\text{easy}) > S(\text{hard})$, matching the observed $2.41 > 1.59$.
- **G2:** the mean of S over the 30 random orders exceeds $S(\text{easy})$, which in turn exceeds $S(\text{hard})$, matching the observed $2.91 > 2.41 > 1.59$.
- **G3**, descriptive and not binding: the Spearman correlation between S and the observed damage across the 30 random orders is positive. Each random order's damage is a single-fit estimate and carries patient and test-set variation, with a standard deviation of 0.70 points across fits, so a weak value is expected either way.

The gate passes only if G1 and G2 both hold. On failure the search is not run, the 120 fits are not submitted, and the outcome is reported as a null. The score is fixed by Equation (2) before the gate runs, and no tuning loop against the gate is permitted: a score adjusted until it reproduces three known damages would no longer be a prediction about them.

4 Results

The gate fails. Of its two binding conditions the first holds and the second does not, so the search was not run and no new fit was spent. Table 1 collects the scores and the damages they are checked against.

G1 passes. The easy-tail order scores 3,239 and the hard-tail order 895, so the score separates the two sorted assignments in the direction their damages take, 2.41 against 1.59 points. The margin is wide: the hard-tail order scores below every one of the 30 random orders, whose scores start at 1,976.

Class order	S	Damage (pp)	Source of the damage
Hard-tail	895	1.59 (1.22 to 1.95)	30 fits, pooled
Random, mean over 30 orders	2,583	2.91 (2.48 to 3.34)	30 fits, pooled
Random, range over 30 orders	1,976 to 3,043	1.53 to 4.69	one fit each
Easy-tail	3,239	2.41 (1.95 to 2.85)	30 fits, pooled

Table 1: The score orders the two sorted assignments correctly but places the easy-tail order above every random order, where the observed damage places it below the random average. Scores are values of Equation (2) at $\rho = 100$; damages are losses of test patient-macro balanced accuracy against the balanced arm, in points, with 95 percent bootstrap intervals where the estimate pools 30 fits. The random-order damages are those of the prevalence curve, one class order per split-draw fit (Queisler, 2026c); the sorted-order damages are those of the tail-assignment study (Queisler, 2026e).

G2 fails. The mean score over the 30 random orders is 2,583, below the easy-tail order’s 3,239, where the gate requires it to lie above. The failure is not marginal. No random order scores as high as the easy-tail order, the largest reaching 3,043, so the score ranks the easy-tail assignment as more damaging than every random assignment observed. The measured damages run the other way: the easy-tail order costs 2.41 points against 2.91 points for the random orders on average, and only 5 of the 30 individual random orders cost less than it does. The score’s most extreme case, among the orders whose damage is known, is one of the milder cases in fact.

G3 is positive. Across the 30 random orders the score and the observed damage correlate at a Spearman coefficient of 0.38 ($p = 0.04$, $n = 30$). The score therefore does carry signal about damage within that family of orders, which makes the failure of G2 sharper rather than softer: the quantity tracks damage among random assignments and inverts it for the one assignment family that is known to lie outside them.

4.1 Why the sorted orders break the score

The inversion follows from how the sorted orders and the score are each built, and does not need a further fit to see. The easy-tail order is defined by sorting the classes on exactly the quantity the score uses as its weight: each class’s balanced-arm recall h_c . Rank and h_c are therefore perfectly monotone in that order, and perfectly monotone in the opposite direction in the hard-tail order, whereas in a random order they are unrelated.

Consider what this does to the terms of Equation (2). In the easy-tail order the classes with the largest h_c occupy the highest rank indices and thus the smallest allocations, so almost every other class sits above them and each of the resulting gaps is multiplied by a large weight. The two factors of every large term, the weight h_c and the log-count advantage of the classes above c , are aligned across the whole order at once. In the hard-tail order the same alignment runs backwards: the large- h_c classes sit at the head with almost nothing above them, their terms are close to zero, and the score collapses to 895, below every random order. A random order achieves neither alignment and lands in between.

The score consequently rewards monotone agreement between recall and rank as such. Ordering the classes by difficulty produces that agreement by construction, in either direction, and so moves the score to an extreme, while the measured damage of both sorted orders is mild, because sorting by difficulty also keeps classes of similar difficulty, and therefore many confused pairs, close together in rank (Queisler, 2026e). What the score was meant to capture, the separation of confused pairs, and what it actually responds to most strongly, the alignment of headroom with rank, come apart precisely on the assignments that are not random.

This matters for the use the score was built for. A search over assignments does not sample the random family; it travels to the extremes of the score, far outside it. The correlation of 0.38

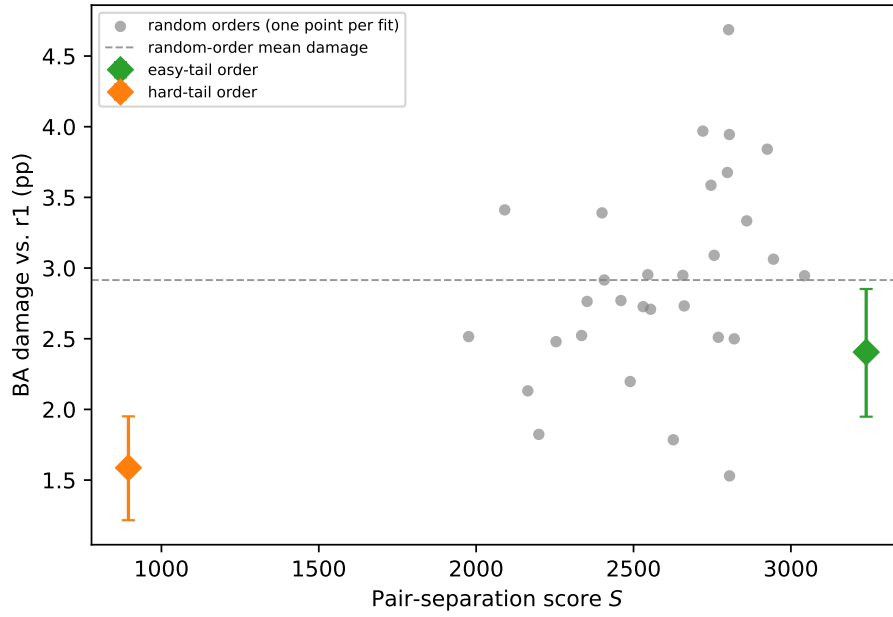


Figure 1: The score tracks damage among random class orders but misplaces the sorted ones. Observed balanced-accuracy damage against the pair-separation score at $\rho = 100$. Grey points: the 30 random class orders of the prevalence curve, one per split-draw fit, with their mean damage as the dashed line. Diamonds with 95 percent bootstrap intervals: the hard-tail order (orange) and the easy-tail order (green), each pooled over 30 fits. The easy-tail order lies to the right of every random order yet below their mean damage, which is the inversion that fails the gate.

within random orders licenses no claim about those extremes, and the single order known to lie outside the random family is the one where the score errs most. Maximizing S would therefore have produced an order chosen for a property that has not been shown to cost accuracy, and the 120 fits would have measured the damage of that order without testing the claim the study set out to test.

5 Discussion

The candidate worst-case assignment named by the tail-assignment study (Queisler, 2026e) is not supported in this form. Written as a single directed sum over confused pairs, weighted by headroom and by log-count gap, it does not rank the three class orders whose damage is known, and the order it ranks as worst is measurably milder than average. The result is a null about the score, reached at the cost of no new fits.

The null is narrower than it may appear, and two things survive it. The pairwise reading of the damage itself is untouched: it rests on per-class recall changes measured under two fitted orders, where rectum adenocarcinoma lost 13 points of recall below colon adenocarcinoma and gained 7 points above it, and where classes whose confusion partners stayed at adjacent ranks barely moved (Queisler, 2026e). Nothing here contradicts that. What fails is the step from that per-pair observation to a scalar summary of a whole assignment. Equally, the possibility that some assignment is markedly worse than a random one remains open; this report shows only that this particular score does not identify it.

The gate itself is the practical lesson. Its three anchor points cost nothing, because all three damages had already been fitted, and it was specified before the score was evaluated. Had the search run first, it would have consumed 120 fits and returned an order with some measured damage, and the resulting number would have been hard to interpret, since a search maximizing a proxy will return an extreme of the proxy whether or not the proxy tracks the target. Checking a proxy against known values before optimizing it is cheap in proportion to what it protects, and it is most informative when the known values include cases from outside the family the proxy was formed on.

A replacement score would need to satisfy two requirements that this one does not. It must not be maximized by monotone agreement between rank and any per-class weight, since assignments with that structure are among the mildest observed here, which suggests scoring each confused pair on its rank gap and direction in a way that does not accumulate across unrelated classes. It must also be checked against anchors that lie outside the random family, and at present only two such anchors exist, the easy-tail and hard-tail orders. Fitting one or two further deliberately constructed orders, cheap at 30 fits each, would strengthen any future gate more than another search would. Such a score would require its own pre-registration; adjusting the present one against the present gate would turn a prediction into a fit.

6 Limitations

The gate rests on three damages, one of them an average, and is a necessary rather than a sufficient check: a score could pass it and still fail to identify the worst assignment, since ranking three known orders correctly says little about the extremes of a space of $30!$ assignments. The two sorted orders are the only available anchors outside the random family, and they are reversals of one another rather than independent constructions, so the evidence that the score misbehaves away from random orders comes from a single pair of related orders.

The damages entering the gate are not measured on a common footing. The sorted-order damages pool 30 fits each with bootstrap intervals, whereas each random-order damage is a single fit whose patient and test-set variation, at a standard deviation of 0.70 points across fits,

attenuates the G3 correlation and would attenuate it even for a score that ranked assignments perfectly. G3 is reported as descriptive for that reason and carries no weight in the pass rule.

The score is evaluated on the $\rho = 100$ count profile. Its ordering of assignments is invariant to the ratio up to a scale factor, so the same order maximizes it at $\rho = 10$, but this invariance is a property of the score rather than a tested property of the damage, and the gate does not test whether the damage ordering itself is stable across ratios. Confusion weights and headroom come from pooled patch-level confusion counts in the balanced arm, which weight classes with many test patches more heavily and do not capture morphological similarity as a pathologist would judge it. All results are conditional on twenty patients per class, frozen Virchow2 features, multinomial logistic regression with regularization selected on validation patient-macro balanced accuracy (Queisler, 2026b), and the three overlapping splits of TCGA-UT.

References

- Yannik Queisler. Class imbalance in histopathology: Skewed prior and thin tail support compound on TCGA-UT. Technical report, TU Berlin, 2026a. Companion TCGA-UT prior-versus-support report, Experiment 27.
- Yannik Queisler. Patient shortage in histopathology: Classifier or coverage limitation? Technical report, TU Berlin, 2026b. Companion patient-shortage report.
- Yannik Queisler. Class imbalance in histopathology: Prevalence alone costs little accuracy on TCGA-UT. Technical report, TU Berlin, 2026c. Companion TCGA-UT prevalence report, Experiment 25.
- Yannik Queisler. Class imbalance in histopathology: Which class sits in the tail matters on BRACS, and easy classes cost most. Technical report, TU Berlin, 2026d. Companion BRACS tail-assignment report, Experiment 29.
- Yannik Queisler. Class imbalance in histopathology: Separated confusion partners, not tail difficulty, set the damage on TCGA-UT. Technical report, TU Berlin, 2026e. Companion TCGA-UT tail-assignment report, Experiment 30.
- Donald B. Rubin. The Bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134, 1981. doi: 10.1214/aos/1176345338.
- Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, George Shaikovski, Neil Tenenholtz, James Hall, Thomas Fuchs, Nicolo Fusi, Siqu Liu, and Kristen Severson. Virchow2: Scaling self-supervised mixed magnification models in pathology. *arXiv preprint arXiv:2408.00738*, 2024. doi: 10.48550/arXiv.2408.00738.